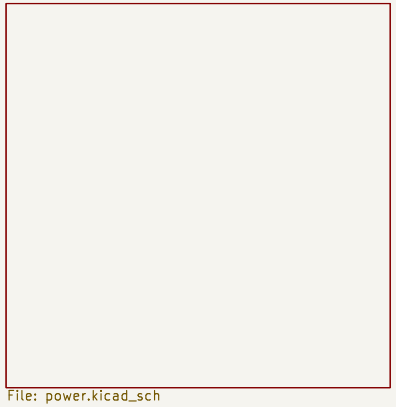
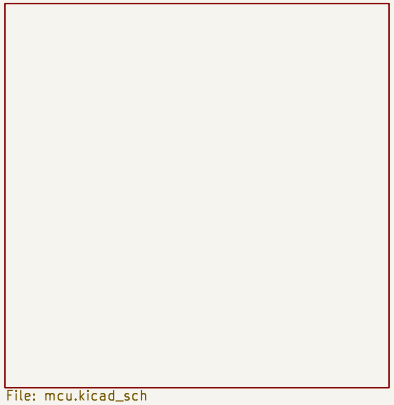
Power



File: power.kicad_sch

MCU



File: mcu.kicad_sch

Driver 1 and matrix



File: matrix1.kicad_sch

Driver 2 and matrix



File: matrix2.kicad_sch

Rails: VBUS -> VSYS 3.0–4.5 V (LED SW lines) -> +3V3 -> +1V1 at the RP2350.
Nets between sheets are global labels; rails are power symbols.
Requirements: design/intent.md. Parts: design/parts/. Decisions: design/decisions.md.

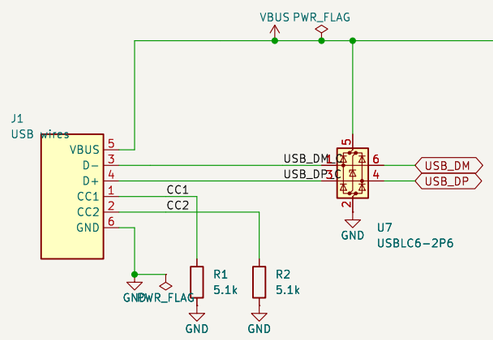
Drawn by scripts/schematic.py
234 RGB LEDs on a 38 mm round board
CERN-OHL-P-2.0

Sheet: /
File: round-pendant.kicad_sch

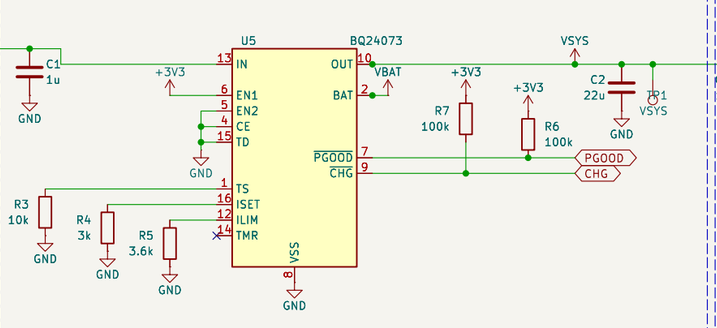
Title: round-pendant: top

Size: A4	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 1/5

USB WIRES, ESD

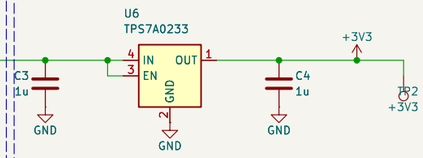


CHARGER AND POWER PATH



USB500: 450−500 mA in, 297 mA charge, ILIM must not float.
TMR open: default 5 h. TS fixed: no NTC.

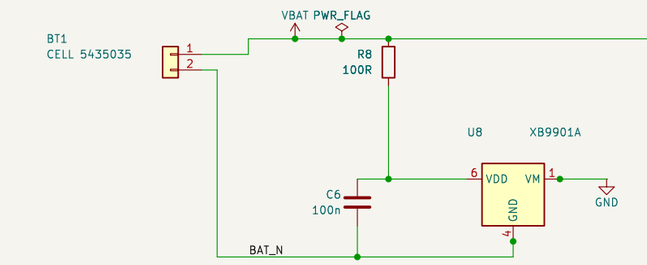
3V3



3V3 only ever follows VSYS down: the driver inputs
stay within VCC + 0.3 V (drivers.md). 150 mA max.

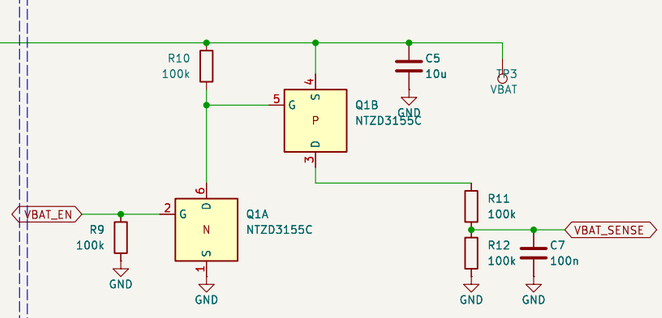
CELL AND PROTECTION

Hand−soldered leads. 610 mAh listed, charged to 4.2 V (battery.md)



BAT_N: cell minus. Pins 4, 5 and the pad. Never pour GND here,
or protection is bypassed (protection.md).

VBAT SENSE



Zero draw off. ADC reads true only while 3V3 regulates.
VBAT >= 3.5 V (intent R22). Wait >= 5 ms after enabling.

Drawn by scripts/schematic.py
USB−C, charger, protection, LDO, VBAT sense
CERN−OHL−P−2.0

Sheet: /Power/
File: power.kicad_sch

Title: round−pendant: power

Size: A3	Date: 2026−11−02	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 2/5

RP2350A, FLASH, CRYSTAL

Regulator: C16 CIN, C18 COUT, L1's polarity and copper follow chips/rp2350.md

100 nF at IOVDD pins 1, 11, 20, 30, 38 (C8–C12), USB_OTP_VDD 53 and QSPI_IOVDD 54 (C13), ADC_AVDD 44 and IOVDD 45 (C14).

C18 COUT at L1; C19 at DVDD 23; C20, C21 at DVDD 6, 39

I2C0 to both drivers, 1 MHz: 12 mA drive, LGC = 1 (drivers.md). Unused GPIOs are left open; firmware clears latched pull-downs before sleep on GPIO 2–5, 8, 9, 12, 13, 26.

Flash pad: float (no internal connection)

VREG_PGND joins GND at one point

ACCELEROMETER (WAKE)

Address 0x19. CS, SAO tied high: ~20 k internal pull-ups would draw 160 uA if strapped low.

BUTTONS

Internal pull-up; wake on low. Hold SW2, tap SW1 for BOOTSEL.

I2C0 PULL-UPS

TEST PADS

CENTRE LED

Average about 1.5 mA per die at full, near a matrix LED's 1.25 mA. Firmware: GPIO high is off; clear pull-downs before sleep.

Drawn by scripts/schematic.py
RP2350A, flash, crystal, accelerometer, buttons
CERN-OHL-P-2.0
Sheet: /MCU/
File: mcu.kicad_sch

Title: round-pendant: mcu

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 3/5

DRIVER 1

Caps at PVCC 15 and 25 and AVCC 14; ground to the chip's GND.
22 uF bulk: edges only. 494 mA scan peak (drivers.md).

VSYS
C30 22u
C31 1u
C32 100n
C33 1u
C34 100n
C35 1u
C36 100n
+3V3
R30 100k
U1
IS31FL3741A
I2C_SDA
I2C_SCL
INTB1
SDB1
C37 100n
R31 100k
R32 30k
ADDR GND: 0x30
30 k: 12.67 mA peak,
1.25 mA average per die

9	SDA	SW1	16 SW1_1
10	SCL	SW2	17 SW1_2
7	INTB	SW3	18 SW1_3
11	SDB	SW4	19 SW1_4
		SW5	20 SW1_5
		SW6	21 SW1_6
13	ISET	SW7	22 SW1_7
8	ADDR	SW8	23 SW1_8
		SW9	24 SW1_9
		CS1	26 CS1_1
		CS2	27 CS1_2
		CS3	28 CS1_3
		CS4	29 CS1_4
		CS5	30 CS1_5
		CS6	31 CS1_6
		CS7	32 CS1_7
		CS8	33 CS1_8
		CS9	34 CS1_9
		CS10	35 CS1_10
		CS11	37 CS1_11
		CS12	38 CS1_12
		CS13	39 CS1_13
		CS14	40 CS1_14
		CS15	41 CS1_15
		CS16	42 CS1_16
		CS17	43 CS1_17
		CS18	44 CS1_18
		CS19	45 CS1_19
		CS20	46 CS1_20
		CS21	47 CS1_21
		CS22	48 CS1_22
		CS23	49 CS1_23
		CS24	50 CS1_24
		CS25	51 CS1_25
		CS26	52 CS1_26
		CS27	53 CS1_27
		CS28	54 CS1_28
		CS29	56 CS1_29
		CS30	57 CS1_30
		CS31	58 CS1_31
		CS32	59 CS1_32
		CS33	60 CS1_33
		CS34	1 CS1_34
		CS35	2 CS1_35
		CS36	3 CS1_36
		CS37	4 CS1_37
		CS38	5 CS1_38
		CS39	6 CS1_39

PVCC 15, PVCC 25, AVCC 14, GND 12

13 x 9 RGB MATRIX: R = CS(3k), G = CS(3k−1), B = CS(3k−2)

SW1_1 SW1_2 SW1_3 SW1_4 SW1_5 SW1_6 SW1_7 SW1_8 SW1_9

CS1_3 CS1_2 CS1_1	D1	D2	D3	D4	D5	D6	D7	D8	D9
CS1_6 CS1_5 CS1_4	D10	D11	D12	D13	D14	D15	D16	D17	D18
CS1_9 CS1_8 CS1_7	D19	D20	D21	D22	D23	D24	D25	D26	D27
CS1_12 CS1_11 CS1_10	D28	D29	D30	D31	D32	D33	D34	D35	D36
CS1_15 CS1_14 CS1_13	D37	D38	D39	D40	D41	D42	D43	D44	D45
CS1_18 CS1_17 CS1_16	D46	D47	D48	D49	D50	D51	D52	D53	D54
CS1_21 CS1_20 CS1_19	D55	D56	D57	D58	D59	D60	D61	D62	D63
CS1_24 CS1_23 CS1_22	D64	D65	D66	D67	D68	D69	D70	D71	D72
CS1_27 CS1_26 CS1_25	D73	D74	D75	D76	D77	D78	D79	D80	D81
CS1_30 CS1_29 CS1_28	D82	D83	D84	D85	D86	D87	D88	D89	D90
CS1_33 CS1_32 CS1_31	D91	D92	D93	D94	D95	D96	D97	D98	D99
CS1_36 CS1_35 CS1_34	D100	D101	D102	D103	D104	D105	D106	D107	D108
CS1_39 CS1_38 CS1_37	D109	D110	D111	D112	D113	D114	D115	D116	D117

Drawn by scripts/schematic.py
IS31FL3741A, 13 x 9 RGB. LED references are cell order only
CERN-OHL-P-2.0
Sheet: /Driver 1 and matrix/
File: matrix1.kicad_sch

Title: round-pendant: driver 1 and matrix

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 4/5

DRIVER 2

Caps at PVCC 15 and 25 and AVCC 14; ground to the chip's GND.
22 uF bulk: edges only. 494 mA scan peak (drivers.md).

VSYS
C40 22u
C41 1u
C42 100n
C43 1u
C44 100n
C45 1u
C46 100n
GND
+3V3
R33 100k
U2
IS31FL3741A
I2C_SDA
I2C_SCL
INTB2
SDB2
C47 100n
R34 100k
R35 30k
SDA 9
SCL 10
INTB 7
SDB 11
ISET 13
ADDR 8
PVCC 15
PVCC 25
AVCC 14
GND 12

ADDR VSYS: 0x33
30 k: 12.67 mA peak,
1.25 mA average per die

SW1 16	SW2_1	CS1 26	CS2_1
SW2 17	SW2_2	CS2 27	CS2_2
SW3 18	SW2_3	CS3 28	CS2_3
SW4 19	SW2_4	CS4 29	CS2_4
SW5 20	SW2_5	CS5 30	CS2_5
SW6 21	SW2_6	CS6 31	CS2_6
SW7 22	SW2_7	CS7 32	CS2_7
SW8 23	SW2_8	CS8 33	CS2_8
SW9 24	SW2_9	CS9 34	CS2_9
		CS10 35	CS2_10
		CS11 37	CS2_11
		CS12 38	CS2_12
		CS13 39	CS2_13
		CS14 40	CS2_14
		CS15 41	CS2_15
		CS16 42	CS2_16
		CS17 43	CS2_17
		CS18 44	CS2_18
		CS19 45	CS2_19
		CS20 46	CS2_20
		CS21 47	CS2_21
		CS22 48	CS2_22
		CS23 49	CS2_23
		CS24 50	CS2_24
		CS25 51	CS2_25
		CS26 52	CS2_26
		CS27 53	CS2_27
		CS28 54	CS2_28
		CS29 56	CS2_29
		CS30 57	CS2_30
		CS31 58	CS2_31
		CS32 59	CS2_32
		CS33 60	CS2_33
		CS34 1	CS2_34
		CS35 2	CS2_35
		CS36 3	CS2_36
		CS37 4	CS2_37
		CS38 5	CS2_38
		CS39 6	CS2_39

13 x 9 RGB MATRIX: R = CS(3k), G = CS(3k−1), B = CS(3k−2)

SW2_1 SW2_2 SW2_3 SW2_4 SW2_5 SW2_6 SW2_7 SW2_8 SW2_9

CS2_3	D118	D119	D120	D121	D122	D123	D124	D125	D126
CS2_2									
CS2_1									
CS2_6	D127	D128	D129	D130	D131	D132	D133	D134	D135
CS2_5									
CS2_4									
CS2_9	D136	D137	D138	D139	D140	D141	D142	D143	D144
CS2_8									
CS2_7									
CS2_12	D145	D146	D147	D148	D149	D150	D151	D152	D153
CS2_11									
CS2_10									
CS2_15	D154	D155	D156	D157	D158	D159	D160	D161	D162
CS2_14									
CS2_13									
CS2_18	D163	D164	D165	D166	D167	D168	D169	D170	D171
CS2_17									
CS2_16									
CS2_21	D172	D173	D174	D175	D176	D177	D178	D179	D180
CS2_20									
CS2_19									
CS2_24	D181	D182	D183	D184	D185	D186	D187	D188	D189
CS2_23									
CS2_22									
CS2_27	D190	D191	D192	D193	D194	D195	D196	D197	D198
CS2_26									
CS2_25									
CS2_30	D199	D200	D201	D202	D203	D204	D205	D206	D207
CS2_29									
CS2_28									
CS2_33	D208	D209	D210	D211	D212	D213	D214	D215	D216
CS2_32									
CS2_31									
CS2_36	D217	D218	D219	D220	D221	D222	D223	D224	D225
CS2_35									
CS2_34									
CS2_39	D226	D227	D228	D229	D230	D231	D232	D233	D234
CS2_38									
CS2_37									

Each cell: RGB, pads 3 1 2 4

Drawn by scripts/schematic.py
IS31FL3741A, 13 x 9 RGB. LED references are cell order only
CERN-OHL-P-2.0
Sheet: /Driver 2 and matrix/
File: matrix2.kicad_sch

Title: round-pendant: driver 2 and matrix

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 9.0.8		Id: 5/5